The explicit FTCS scheme for the Black–Scholes PDE

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1 Setting

This note implements the Forward-Time, Centred-Space (FTCS) finite- difference pricer for European calls and puts. We use the constant-coefficient transformed Black–Scholes equation derived in the foundations note (*Foundations of the finite-difference solution of the Black-Scholes PDE*),

$$\frac{\partial V}{\partial \tau} \;=\; \frac{\sigma^2}{2}\frac{\partial^2 V}{\partial x^2} + \mu\,\frac{\partial V}{\partial x} - r\,V, \qquad \mu = r - \tfrac{\sigma^2}{2}, \tag{1}$$

with $x = \ln(S/K)$, $\tau = T - t$, on the truncated domain $[x_{\min}, x_{\max}] \times [0, T]$ with Dirichlet boundary conditions specified there. The deliverable is a pricer, validated against the Black–Scholes closed form and analysed in terms of stability and convergence.

2 The FTCS discretisation

Let $V_j^n \approx V(x_j, \tau_n)$ on the uniform grid $x_j = x_{\min} + j\,\Delta x$, $\tau_n = n\,\Delta\tau$. Substituting the centred space stencils and the forward-time stencil into (1):

$$\frac{V_j^{n+1} - V_j^n}{\Delta\tau} \;=\; \frac{\sigma^2}{2}\,\frac{V_{j+1}^n - 2\,V_j^n + V_{j-1}^n}{\Delta x^2} + \mu\,\frac{V_{j+1}^n - V_{j-1}^n}{2\,\Delta x} - r\,V_j^n.$$

All spatial differences are evaluated at the *old* time level n. This is what makes the scheme explicit: the new-time value V_j^{n+1} is given directly as a linear combination of three old-time values, without solving any system.

Multiplying by $\Delta\tau$ and isolating V_j^{n+1} on the left,

$$\boxed{V_j^{n+1} \;=\; a_-\,V_{j-1}^n \;+\; a_0\,V_j^n \;+\; a_+\,V_{j+1}^n} \tag{2}$$

with constant coefficients

$$a_- = \alpha - \frac{\mu\,\Delta\tau}{2\,\Delta x}, \quad a_0 = 1 - 2\alpha - r\,\Delta\tau, \quad a_+ = \alpha + \frac{\mu\,\Delta\tau}{2\,\Delta x}, \tag{3}$$

where $\alpha = \frac{\sigma^2}{2}\,\Delta\tau/\Delta x^2$ is the Fourier number of the foundations note. The coefficients do not depend on j or n, by virtue of the constant-coefficient structure of (1): this is the structural pay-off of the logarithmic transformation.

3 Probabilistic interpretation

The three coefficients sum to

$$a_- + a_0 + a_+ \;=\; 1 - r\,\Delta\tau.$$

This identity has a clean reading: if a_-, a_0, a_+ are non-negative, define

$$p_- = \frac{a_-}{1 - r\Delta\tau}, \quad p_0 = \frac{a_0}{1 - r\Delta\tau}, \quad p_+ = \frac{a_+}{1 - r\Delta\tau},$$

so that $p_-, p_0, p_+ \geq 0$ and $p_- + p_0 + p_+ = 1$. Then (2) is exactly

$$V_j^{n+1} \;=\; (1 - r\,\Delta\tau)\left(p_-\,V_{j-1}^n + p_0\,V_j^n + p_+\,V_{j+1}^n\right) \;\approx\; e^{-r\Delta\tau}\,\mathbb{E}^{\mathbb{Q}}\Big[\,V(x_j + \xi\,\Delta x,\,\tau_n)\,\Big], \quad (4)$$

where $\xi \in \{-1, 0, +1\}$ with $\mathbb{Q}$-probabilities (p_-, p_0, p_+). The factor $1 - r\,\Delta\tau$ is the forward-Euler approximation to the discount factor $e^{-r\Delta\tau}$.

Equation (4) says that FTCS is the discretisation of the risk-neutral pricing identity

$$V(x_j, \tau_{n+1}) \;=\; e^{-r\Delta\tau}\,\mathbb{E}^{\mathbb{Q}}\big[V(X_{n+1}, \tau_n)\,\big|\,X_n = x_j\big]$$

on a discrete state space, with $X_{n+1} - X_n$ taking the values $\pm\Delta x, 0$. This is exactly the structure of a trinomial tree. The trinomial-tree note develops this connection: the Kamrad–Ritchken tree is, after a precise reparametrisation, FTCS in disguise.

The reading also highlights why the non-negativity of the coefficients matters: if any $p_\xi < 0$, the operation loses its interpretation as a discrete expectation, and the scheme can amplify oscillations near the strike kink in the payoff.

4 Non-negativity of the coefficients

Three conditions, one per coefficient.

Central coefficient. $a_0 \geq 0$ requires

$$\alpha \;\leq\; \frac{1 - r\,\Delta\tau}{2}. \quad (5)$$

For $r\,\Delta\tau \ll 1$ (always true in this project: $r \leq 0.1$ and $\Delta\tau \lesssim 0.02$, so $r\,\Delta\tau \leq 2 \times 10^{-3}$), (5) is essentially the CFL condition $\alpha \leq 1/2$ of the foundations note.

Lateral coefficients. $a_\pm \geq 0$ requires $\big|\,\mu\,\Delta\tau/(2\Delta x)\big| \leq \alpha$, equivalently

$$\frac{|\mu|\,\Delta x}{\sigma^2} \;\leq\; 1. \quad (6)$$

The dimensionless ratio $\mathrm{Pe}_h := |\mu|\Delta x/\sigma^2$ is the *grid Péclet number*: it measures the relative weight of convection to diffusion at the scale of one grid cell. For the parameter range of this project ($\sigma \in [0.1, 0.4]$, $|\mu| \leq 0.1$, $\Delta x \leq 0.1$), $\mathrm{Pe}_h \leq 1$ is satisfied, with equality at the corner $\sigma = 0.1$, $|\mu| = 0.1$, $\Delta x = 0.1$. The grids used in the validation and benchmark scripts have $\mathrm{Pe}_h \leq 0.025$.

Remark 4.1 (Convection-dominated regimes)**.** *When* $\mathrm{Pe}_h > 1$ *(extreme dividends, very small volatility, very coarse grid), centred differences for the convective term produce one negative lateral coefficient, the probabilistic reading breaks, and the scheme exhibits localised oscillations near the strike kink. The standard remedy is an* upwind *discretisation of the convective term: replace* $(V_{j+1} - V_{j-1})/(2\Delta x)$ *by* $(V_j - V_{j-1})/\Delta x$ *if* $\mu > 0$ *or* $(V_{j+1} - V_j)/\Delta x$ *if* $\mu < 0$*. This restores* L^∞ *stability at the price of one order of accuracy in space (*$O(\Delta x)$ *instead of* $O(\Delta x^2)$*). Out of scope here.*

Summary. The probability-positive (monotone) regime is

$$\boxed{\alpha \le \tfrac{1}{2} \text{ and } \mathrm{Pe}_h \le 1.} \tag{7}$$

In this regime the scheme is monotone, and monotonicity implies stability in the uniform L^∞ norm:

$$\max_j |V_j^{n+1}| \;\le\; (1 - r\Delta\tau) \max_j |V_j^n| \;\le\; \max_j |V_j^n|.$$

This is a strictly stronger result than the L^2 stability of the next section, and it is the practical reason the prices computed by FTCS *look smooth*: oscillations cannot grow.

5 Von Neumann stability analysis

The von Neumann analysis of (2) for the full convection-diffusion-reaction equation supplements the previous section with a tight necessary-and-sufficient condition for L^2 stability.

Substituting the Fourier ansatz $V_j^n = g^n e^{ij\theta}$ into (2) and using $e^{i\theta} + e^{-i\theta} = 2 - 4\sin^2(\theta/2)$, $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$, the amplification factor is

$$g(\theta) \;=\; (1 - r\,\Delta\tau) - 4\,\alpha\,\sin^2\tfrac{\theta}{2} + i\,\beta\,\sin\theta, \qquad \beta \;:=\; \frac{\mu\,\Delta\tau}{\Delta x}. \tag{8}$$

(β is the unsymmetrised Courant number; it differs from the ν of the foundations note by a factor of 1/2.) The factor is complex: the real part is the diffusive contribution shifted by the reactive term; the imaginary part is the convective contribution.

Proposition 5.1 (Stability of FTCS). *Set $r = 0$ for clarity (the effect of the reactive term is discussed at the end of this section). The condition $|g(\theta)|^2 \le 1$ for all $\theta \in [-\pi, \pi]$ is equivalent to*

$$\alpha \;\le\; \tfrac{1}{2} \qquad \textit{and} \qquad \beta^2 \;\le\; 2\,\alpha. \tag{9}$$

Proof. With $s := \sin^2(\theta/2) \in [0, 1]$, $\sin^2\theta = 4s(1 - s)$. Then

$$|g|^2 = (1 - 4\alpha s)^2 + 4\beta^2 s(1 - s),$$

and after expansion,

$$|g|^2 - 1 = s\left[(16\alpha^2 - 4\beta^2)\,s + (4\beta^2 - 8\alpha)\right].$$

The bracket is linear in s. The condition $|g|^2 \le 1$ for all $s \in (0, 1]$ is equivalent to the bracket being non-positive at both endpoints $s = 0$ and $s = 1$ (linear function on a compact interval attains its extrema at the endpoints). At $s = 0$: $4\beta^2 - 8\alpha \le 0 \Leftrightarrow \beta^2 \le 2\alpha$. At $s = 1$: $16\alpha^2 - 8\alpha \le 0 \Leftrightarrow \alpha \le 1/2$. □ □

The two conditions are not equivalent to monotonicity. The von Neumann conditions (9) are about L^2 stability; the conditions (7) are about monotonicity. The relationship between them is:

- Both conditions share $\alpha \le 1/2$.
- The von Neumann condition $\beta^2 \le 2\alpha$ rewrites as $\mu^2\Delta\tau \le \sigma^2$, a constraint on $\Delta\tau$. For typical parameters ($\sigma = 0.2$, $|\mu| \le 0.1$): $\Delta\tau \le \sigma^2/\mu^2$, which is at least 4 years. Always satisfied.
- The probability-positive condition $\mathrm{Pe}_h \le 1$ rewrites as $|\mu|\Delta x \le \sigma^2$, a constraint on Δx.

The two are different conditions on different variables. The probability-positive constraint is strictly stronger: it implies L^2 stability via monotonicity, but the converse fails. In our parameter range both conditions are satisfied; in convection-dominated regimes, they decouple, and the von Neumann condition can hold without monotonicity (giving stable but oscillatory schemes).

What dominates in practice. The constraint that bites is $\alpha \le 1/2$. The Péclet condition $\mathrm{Pe}_h \le 1$ is a constraint on Δx that is comfortably met on the grids used here ($\mathrm{Pe}_h \le 0.025$, Section 4). The reactive contribution shifts g by $-r\Delta\tau$: it damps the modes with $\mathrm{Re}\, g > 0$ but increases $|g|$ where $\mathrm{Re}\, g < 0$. At $\theta = \pi$, $g = 1 - 4\alpha - r\Delta\tau$, so the strict criterion $|g| \le 1$ requires $\alpha \le \frac{1}{2} - \frac{r\Delta\tau}{4}$, a small tightening that the monotonicity bound (5) already implies. Under the generalised von Neumann criterion $|g| \le 1 + O(\Delta\tau)$ the reactive term never affects stability.

6 The time-marching algorithm

In pseudocode, the inner loop is

```
V[j] for j = 0..N  <-  initial-condition vector at tau = 0
for n = 0, 1, ..., M-1:
    V_new[0] = bc_lower[n+1]                          # Dirichlet
    V_new[N] = bc_upper[n+1]                          # Dirichlet
    for j = 1, ..., N-1:                              # interior nodes
        V_new[j] = a_minus * V[j-1]
                 + a_zero  * V[j]
                 + a_plus  * V[j+1]
    swap(V, V_new)
return V                                              # V at tau = T
```

Three implementation points worth explicit comment.

Double buffer. The update of node j at level $n+1$ uses V^n_{j-1} and V^n_{j+1}. If the implementation overwrites V_j in place, the value of V_j used by the update of V_{j+1} will be the new one rather than the old, breaking the scheme. Two buffers V and V_{new} are needed, swapped after each step.

Boundary conditions are not stencil applications. The boundary nodes $j = 0$ and $j = N$ are fixed by the Dirichlet conditions of the foundations note; the stencil is applied only to interior nodes $j = 1, \ldots, N-1$. Trying to apply the stencil at $j = 0$ would require a phantom node $j = -1$ outside the domain.

Recovery of the price at S_0. The grid is in $x = \ln(S/K)$. The pricer must return $V(S_0, 0)$ at the user's spot S_0, which is generally not on a grid node. The V array at $\tau = T$ contains V^M_j for $j = 0, \ldots, N$, and the price is recovered by interpolation at $x_0 = \ln(S_0/K)$. Linear interpolation is enough: it adds an $O(\Delta x^2)$ error consistent with the spatial discretisation order.

7 Convergence and computational cost

Local truncation error. By Taylor expansion of every stencil,

$$\tau^n_j \;=\; O(\Delta\tau) + O(\Delta x^2).$$

First order in $\Delta\tau$ from the forward-Euler step; second order in Δx from the centred space stencils.

Global error and convergence rate. By the Lax equivalence theorem (foundations note, §9), consistency plus stability gives convergence,

$$\left\| V_h - V \right\|_\infty \;=\; O(\Delta\tau) + O(\Delta x^2).$$

Under the CFL constraint $\Delta\tau \le \Delta x^2/\sigma^2$, both contributions are $O(\Delta x^2)$, so the scheme converges quadratically in Δx. There is no parameter regime in which FTCS achieves better than $O(\Delta x^2)$ convergence: the temporal order is locked at 1, and the spatial order is locked at 2, but the CFL constraint balances the two automatically.

Computational cost. Each time step does $O(N)$ work (one stencil application per interior node). The number of time steps is, by CFL,

$$M \;\ge\; \frac{T\,\sigma^2}{\Delta x^2} \;\sim\; N^2.$$

Total work: $O(N \cdot M) = O(N^3)$. Doubling N multiplies the cost by 8. This N^3 scaling is the price tag of explicit time stepping; BTCS and Crank–Nicolson sidestep it by allowing $\Delta\tau \sim \Delta x$ at the cost of solving a tridiagonal system per step.

8 Implementation outline

The Python module `quantlib.ftcs` exposes `ftcs_european_call` and `ftcs_european_put`, both with signature `(S, K, r, sigma, T, *, N, M, n_sigma=4.0)`. The C++ counterpart in the `quant::pde` namespace has the same arguments. Both validate that $\alpha \le 1/2$ at construction and raise an error otherwise (Python: `ValueError`; C++: `std::invalid_argument`). The validation is strict and not configurable: the pricer will not silently produce nonsense from a violated CFL.

The validation tests (Python: `validate_ftcs.py`; C++: `test_ftcs.cpp`) check four properties:

1. agreement with the Black–Scholes closed form on a grid of parameters, within a fixed absolute tolerance of 5×10^{-3};
2. the empirical convergence rate, by running on (N, M) and $(2N, 4M)$ and checking that the error ratio is close to 4;
3. put-call parity: $C - P = S - Ke^{-rT}$ at every spot, to within twice the discretisation tolerance;
4. that violating CFL ($\alpha > 1/2$) raises the appropriate exception.

A final diagnostic in `validate_ftcs.py` disables the CFL check and runs the scheme at $\alpha \approx 1.56$ ($N = 100$, $M = 50$) to show the sawtooth instability mode: the error at the highest grid frequency grows without bound, as predicted by the von Neumann analysis. After the 50 steps, $\max_j |V_j| \approx 3 \times 10^{34}$ and V changes sign 99 times across the 101 nodes.

References

References

[1] D. Tavella and C. Randall. *Pricing Financial Instruments: The Finite Difference Method.* Wiley, 2000. Chapter 2 (Analysis of Finite Difference Methods).

[2] R. Seydel. *Tools for Computational Finance.* 6th ed., Springer, 2017. Chapter 4 (Standard Methods for Standard Options).

[3] J. C. Strikwerda. *Finite Difference Schemes and Partial Differential Equations.* 2nd ed., SIAM, 2004. Chapter 2 (von Neumann analysis) and Chapter 6 (parabolic equations).

[4] K. W. Morton and D. F. Mayers. *Numerical Solution of Partial Differential Equations.* 2nd ed., CUP, 2005. §2.4–2.7 (the explicit scheme, its truncation error, convergence and Fourier analysis), §5.5 (stability and the Lax equivalence theorem).

[5] J. Hull. *Options, Futures, and Other Derivatives.* 10th ed., Pearson, 2018. Chapter 21 (FTCS $\leftrightarrow$ trinomial tree relation, from the tree side).

[6] P. P. Boyle. *Option valuation using a three-jump process.* International Options Journal 3, 7–12, 1986. Introduces the trinomial tree, with a stretch parameter.

[7] B. Kamrad and P. Ritchken. *Multinomial approximating models for options with k state variables.* Management Science 37(12), 1640–1652, 1991. Generalises Boyle's trinomial tree; see the trinomial-tree note.